

SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING
Academic Year: 2024-2025

Program : BTI/B.Tech/MBA Tech	Year :	IV /II
Stream : Data Science	Semester :	VIII/IV
Subject : Stochastic Process & Applications (STPA)	Time :	3 hrs (10:00AM to 1:00PM)
Date : 28 April 2025	No. of Pages :	3
Marks : 100		

**Final Examination 2024-25/
Re-Examination 2022-23/2024-25**

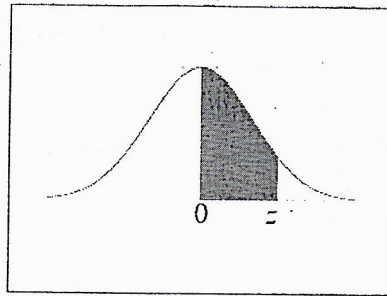
Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Solve **Total FIVE** questions.
- 2) Question number **01** is compulsory.
- 3) All questions carry equal marks.
- 4) Answer to each new question to be started on a fresh page.
- 5) Figures in brackets on the right-hand side indicate full marks.
- 6) Assume Suitable data if necessary.
- 7) Scientific Calculator & Statistical Table is allowed to use.

Q1.	Answer in Detail	[20]
a.	Define first and second-order statistics of a random process.	[5]
b.	The lifetime of certain brand of electric bulb may be considered a random variable with mean 1200 and SD 250 hours. Using CLT to find the probability that the average lifetime of 60 bulbs exceeds 1250 hours.	[5]
c.	Consider a Markov Chain with states $\{0, 1\}$ and the transition probability matrix is $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Is the state 0 is periodic, if yes, find the period?	[5]
d.	If the autocorrelation of a random process is given by $R(\tau) = \begin{cases} \tau & \tau \leq 1 \\ 0 & \tau > 1 \end{cases}$ then find the spectral density function $S_{xx}(\omega)$.	[5]
Q2.	Answer in Detail	[20]
a.	Find the mean, mean square and variance of a random process whose autocorrelation function is given by $R_{XX}(\tau) = 18 + \frac{2}{(6 + \tau^2)}$	[6]
b.	Consider the outcomes of the tossing of fair coins are described by the following function of time $X(t) = \begin{cases} \sin \pi t, & \text{If head appears.} \\ 2t, & \text{If tail appears.} \end{cases}$ Where $t > 0$. Describe the random process.	[6]
c.	If X denotes the outcome when a fair die is tossed. Then obtain the MGF $M_X(t)$ and henceforth figure out the corresponding mean and variance of the distribution.	[8]
Q3.	Answer in Detail	[20]
a.	Suppose that the average number of telephone calls arriving at a police station is 30 calls per hour. What is the probability that (i) No call will arrive in a 3-minute period. (ii) More than 4 calls will arrive in a 5-minute period?	[6]
b.	Two random processes are defined by $X(t) = \cos(\omega t + \Theta)$ and $Y(t) = \sin(\omega t + \Theta)$, where ω is constant and $\Theta \sim U[-\pi, \pi]$. Find the cross correlation $R_{XY}(\tau)$.	[6]
c.	Show that the random process given by $X(t) = X_1(t) \cos t + X_2(t) \sin t$, where X_1, X_2 are independent discrete random variables assuming values -1 and 2 with probabilities $\frac{2}{3}$ and $\frac{1}{3}$ is wide sense stationary but not strict sense stationary.	[8]

Q4.	Answer in Detail	[20]																		
a.	If a random process is given by $X(t) = A \cos(2\pi t + Y)$, Where A is a constant and Y is discrete random variable with $P(Y = 0) = P\left(Y = \frac{\pi}{2}\right) = \frac{1}{2}$, then find the mean and autocorrelation function.	[6]																		
b.	The transition probability matrix P of a Markov Chain $\{X_n\}$ where $n \in \mathbb{N}$, with the three states, numbered 1, 2, and 3 given by $P = \begin{bmatrix} 0.4 & 0.6 & 0.3 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.7 & 0.2 \end{bmatrix}$, with the initial distribution is $P^{(0)} = [0.6 \quad 0.3 \quad 0.1]$ Then find the probabilities of the following states: (a) If today is Wednesday and the weather is in state 2 what is the probability that the weather is in state 3 on Thursday and state 1 on Friday? (b) What is the probability that the weather on Friday is in state 1, given that it is in state 2 on Wednesday?	[6]																		
c.	A continuous function of a random variable X has the p.d.f $f(x) = kx^2 e^{-x}$ where $x \geq 0$. Then find (i) k , (ii) Mean (iii) Variance.	[8]																		
Q5.	Answer in Detail	[20]																		
a.	If the power spectral density $S_{xx}(\omega)$ of a random process is given by $S_{xx}(\omega) = \frac{10\omega^2 + 35}{\omega^4 + 13\omega^2 + 36}$ then find the autocorrelation function $R_{xx}(\tau)$ and hence obtain the average power of the process.	[6]																		
b.	If $\{X(t)\}$ is a Gaussian process with $\mu(t) = 10$ and the cross correlation function is $C(t_1, t_2) = 4e^{-0.2 t_1 - t_2 }$ Then find $P(X(8) \leq 3)$ and $P(X(8) - X(3)) \leq 3$.	[6]																		
c.	Suppose in a gambling game a gambler wins and losses rupees 1 with probability 0.45 and 0.55 respectively at any stage. He starts the game with 1 rupee and quits it when either he has earned rupees 4 or he is bankrupt. Define the process as a Markov chain and write the transition matrix. Also obtain the transition probability of sequences $\{12343210\}$ and $\{234345\}$.	[8]																		
Q6.	Answer in Detail	[20]																		
a.	Consider the DRV X is associated with the following PMF <table><tr><td>X</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>$P_X(x)$</td><td>$\frac{1}{8}$</td><td>$\frac{3}{4}$</td><td>$\frac{1}{8}$</td></tr></table> Then evaluate $P\{ X - \mu \geq 2\sigma\}$ and compare the result with the upper bound of Chebyshev's inequality.	X	-1	0	1	$P_X(x)$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{8}$	[6]										
X	-1	0	1																	
$P_X(x)$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{8}$																	
b.	The autocorrelation function of a stationary random process is given by $R_X(\tau) = Ae^{-\alpha \tau }$. Find the second moment of the random variable $Y = X(6) - X(3)$.	[6]																		
c.	Prove that the random process $X(t) = A \cos(\omega t + \Theta)$ is mean ergodic and correlation-ergodic, where A and ω are constants and $\Theta \sim U[0, 2\pi]$.	[8]																		
Q7.	Answer in Detail	[20]																		
a.	A random variable X has the following probability distribution <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>$P_X(x)$</td><td>0</td><td>k</td><td>$2k$</td><td>$2k$</td><td>$3k$</td><td>k^2</td><td>$2k$</td><td>$7k^2 + k$</td></tr></table> Then find (i) k (ii) $P(0 < X < 5)$ (iii) The CDF of X .	X	0	1	2	3	4	5	6	7	$P_X(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k$	$7k^2 + k$	[6]
X	0	1	2	3	4	5	6	7												
$P_X(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k$	$7k^2 + k$												
b.	A sine wave random process is given by $X(t) = A \sin \omega_0 t$ where $A \sim U[-1, 1]$ and ω_0 is a constant. Check whether $\{X(t)\}$ is WSS.	[6]																		
c.	Find the spectral density function $S_{xx}(\omega)$ where $R(\tau) = e^{-a \tau } \cos b\tau$.	[8]																		

Standard Normal Distribution Table

[illegible]

